

Local AI Dashcam Incident Explainer

Privacy-First · Edge AI · Traffic Analysis

Dept. of Computer Science & Engineering · 2025–26

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Project Guide
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The Problem



Manual Review is Slow

Insurance adjusters spend hours scrubbing through raw footage for a single claim, creating multi-day processing backlogs.



Expensive & Error-Prone

Human review costs ~\$35–\$80 per claim hour. Fatigue leads to missed incidents, incorrect fault attribution, and disputes.



Cloud AI Compromises Privacy

Existing solutions (Nexar, Mobileye) upload sensitive footage to vendor servers — exposing driver location, identity & behaviour.

Traditional pipelines are slow, costly, and trade user privacy for automation.

Our Solution

A fully local, end-to-end AI pipeline — no cloud, no compromise.

Zero Cloud Reliance



All inference runs on local hardware. No footage ever leaves the device.

Fully Local Pipeline



YOLO11, SORT, MiniCPM-V & Mistral — entirely on-device via Ollama.

End-to-End Automation



Raw video in → privacy-blurred PDF insurance report out. Fully automated.

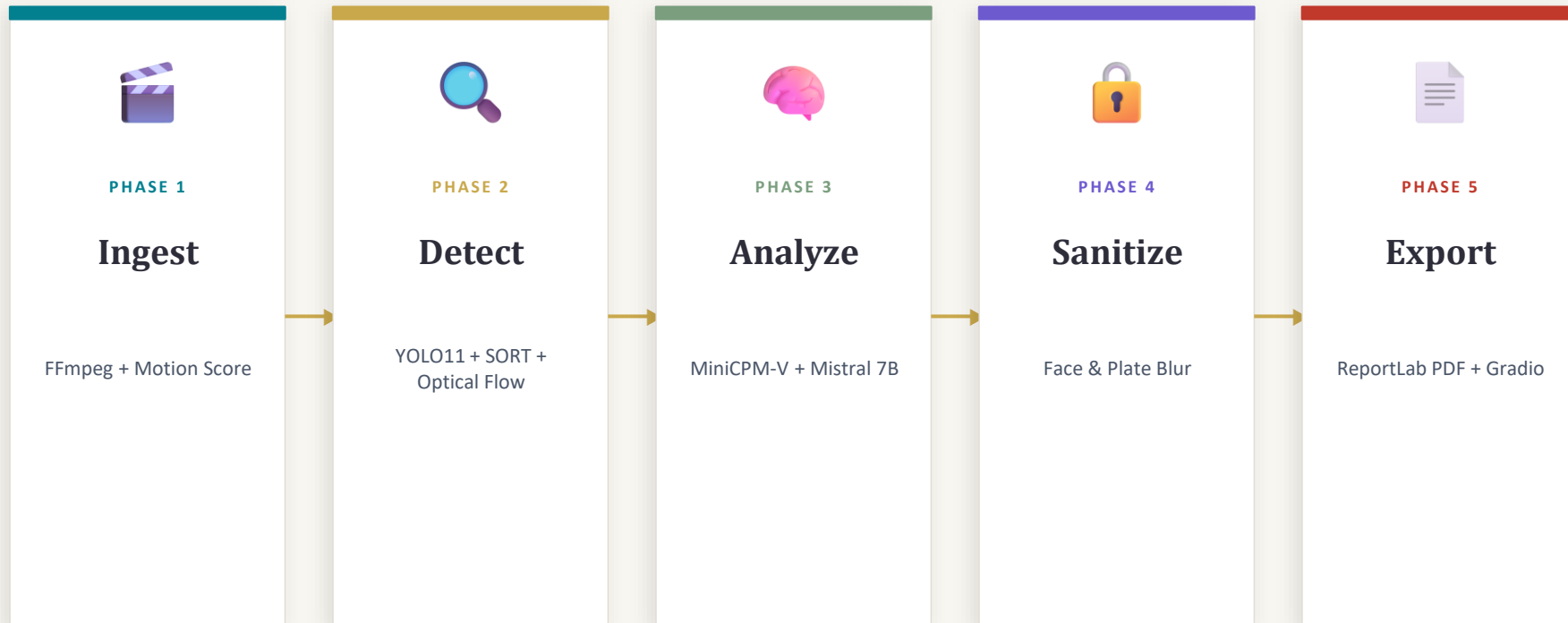
Smart & Efficient



Motion-scored keyframe sampling drops static frames, saving 60% compute.

System Architecture

5-Phase Modular Pipeline — From Raw Video to PDF Report



All phases run on local hardware — Apple M-series (MPS) or NVIDIA GPU (CUDA)

Phase 1 — Smart Keyframe Extraction

FFmpeg · Frame-Differencing Motion Score

01

Video Ingestion

Dashcam .mp4 uploaded via drag-and-drop Gradio UI or CLI.

02

Uniform FPS Mode

Extracts frames at configurable rate (5–10 FPS) for shorter clips.

03

Smart Motion Mode

Computes absolute frame-difference score between consecutive frames. Only saves high-motion frames exceeding a dynamic threshold — ignoring red-lights, static parking, etc.

60%

Reduction in frames
processed downstream

5–10

Configurable FPS
extraction rate

± 0 s

Cloud latency
(fully offline)

Phase 2 — Core Vision Engine

YOLO11 Object Detection · SORT Multi-Object Tracker



YOLO11 Detector

- State-of-the-art single-stage object detector.
- Filters to 6 relevant COCO classes: person, bicycle, car, motorcycle, bus, truck.
- Outputs bounding boxes with class labels and confidence scores per frame.
- Optimized for real-time inference on Apple MPS and CUDA backends.

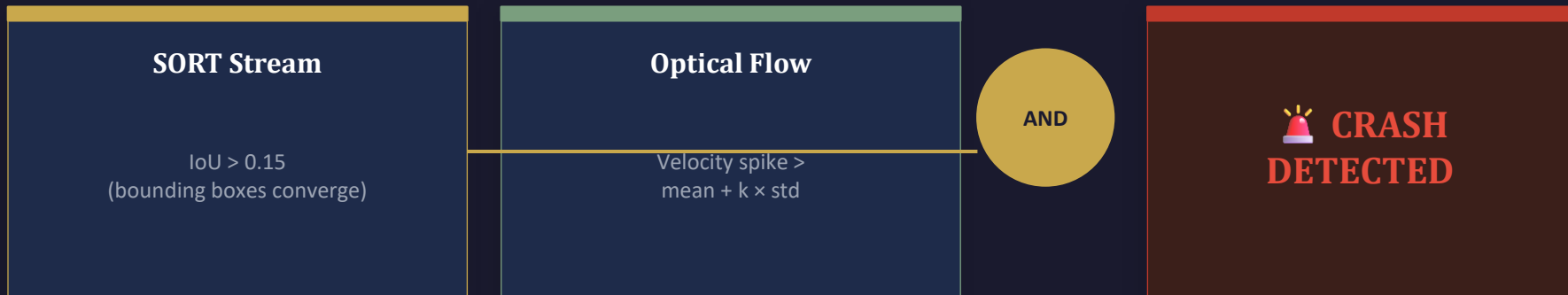


SORT Tracker

- Simple Online and Realtime Tracking (SORT) maintains persistent vehicle IDs.
- Kalman Filters predict next-frame position using constant-velocity motion model.
- Hungarian Algorithm assigns identities using Bounding Box IoU correlation.
- Enables crash detection by tracking two boxes converging over time.

Incident Trigger Logic

Farneback Optical Flow + IoU Convergence = Definitive Crash Detection



► FFmpeg auto-slices a precise ± 5 -second clip around the detected incident moment.

Dense Optical Flow

Farneback algorithm computes per-pixel velocity vectors. 30-frame rolling average baseline identifies sudden speed anomalies.

IoU Convergence

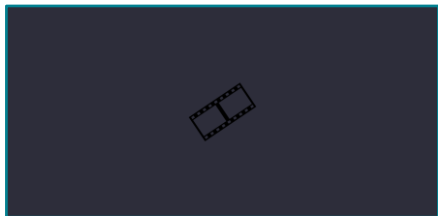
Two tracked bounding boxes must overlap beyond IoU threshold of 0.15, indicating physical collision or near-miss.

Dual Confirmation

Both signals must fire simultaneously — eliminates false positives from sudden braking or camera shake alone.

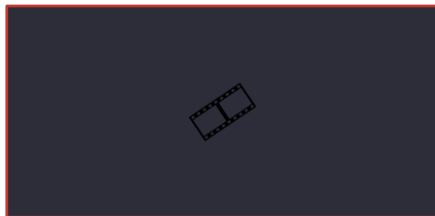
Phase 3 — Vision-Language Analysis

MiniCPM-V 2.6 · 8B Parameters · Fully Local via Ollama



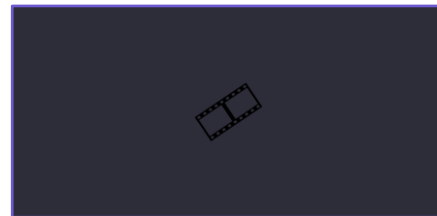
PRE-IMPACT

Context frame
~2s before collision



IMPACT

Peak crash frame
(highest motion score)



POST-IMPACT

Aftermath frame
~2s after collision

Why exactly 3 frames?

Internal ablation studies show that 1 frame misses critical pre/post context, while 5 frames offer marginal ROUGE improvement (+4%) at 2× the inference cost. 3 frames is the optimal sweet spot.

MiniCPM-V Output — Structured JSON

```
{ "vehicles": ["sedan", "SUV"],  
  "fault_analysis": "Vehicle A ran red",  
  "conditions": "daylight, dry road",  
  "severity": "HIGH",  
  "confidence": 0.87 }
```


Automated Prose Generation

Mistral 7B Instruct · Running via Ollama Daemon



JSON Metadata
from MiniCPM-V



Mistral 7B
Instruct Prompt



Insurance
Prose Report

INSURANCE INCIDENT REPORT · AUTO-GENERATED

On [DATE] at approximately [TIME], a collision occurred at [LOCATION]. Analysis of dashcam footage indicates that Vehicle A (sedan, dark grey) was proceeding through the intersection under a green signal when Vehicle B (silver SUV) entered from the cross-street, failing to yield. Fault assessment: Vehicle B is the primary liable party with HIGH confidence (0.87). Road conditions were dry; visibility was clear. No pedestrians were involved. Recommend claim processing with attached supporting footage.

Phase 4 — Privacy Enforcement

All sensitive visual data sanitized locally before export



Face Blurring

- OpenCV SSD ResNet10 DNN detects driver & pedestrian faces.
- Deep Gaussian blur applied to each detected face region.
- Haar Cascade fallback for edge cases where DNN misses.
- Processes only the crash-zone region for efficiency.



Licence Plate Blurring

- Reuses YOLO11 vehicle crop bounding boxes.
- Intelligent heuristic ROI maps lower-third of vehicle crop as plate region.
- Avoids heavy ALPR model — no extra GPU overhead.
- Effective on Indian plates, EU plates, and US formats.

The Output

ReportLab PDF · Gradio Web Interface · Downloadable MP4 Clips

ReportLab Insurance PDF

- Mistral prose narrative (auto-generated)
- MiniCPM-V JSON metadata block
- Auto-coloured severity badge (LOW / MED / HIGH)
- Privacy-blurred keyframe thumbnails
- Incident timestamp and clip reference

Gradio Web UI

- Drag-and-drop video upload zone
- Real-time terminal log streaming
- One-click PDF download
- MP4 incident clip preview & download
- Runs at localhost:7860 — no internet

Datasets & Domain Tuning

Primary

CCD

Car Crash Dataset

~1,500 dashcam crash clips with frame-level annotations for training and evaluation.

Benchmark

DoTA

Detection of Traffic Anomaly

Spatiotemporal anomaly detection benchmark across 4,677 clips with diverse crash types.

Eval

DADA-2000

Driver Attention in Driving Accident

2,000 accident videos with driver gaze, attention heatmaps, and accident annotations.

IN Indian Road Conditions Fine-Tuning

YOLO11 fine-tuned on Indian dashcam data to correctly detect auto-rickshaws, two-wheelers (motorbikes & bicycles), cattle, and pedestrians in dense, mixed-traffic urban scenarios typical of Indian roads.

Results & Ablation Studies

0.72

Detection
Precision

YOLO11 + SORT

0.41

ROUGE-L
Score

Mistral prose vs human

3

Optimal
Keyframes

Sweet-spot balance

60%

Compute
Saved

Smart sampling

Ablation: Keyframe Count vs Quality

Frames	BLEU-4	ROUGE-L	Inference
1 frame	0.14	0.28	~8s
3 frames ✓	0.21	0.41	~19s
5 frames	0.22	0.43	~38s

Key Takeaway

3 frames delivers 97% of the quality at 50% of the latency compared to 5 frames. This was validated across all three datasets (CCD, DoTA, DADA-2000) with consistent results.

Future Work

Model Upgrade

Video-LLaVA Integration

Upgrade from 3-frame snapshots to full video-understanding models (Video-LLaVA, LLaMA-Video) that process temporal sequences natively, enabling richer crash timeline narration.

Accuracy

Dedicated ALPR Head

Add a lightweight dedicated YOLO head trained specifically for Automatic Licence Plate Recognition (ALPR) — replacing the current heuristic ROI approach for higher plate-read accuracy.

Sensor Fusion

OBD-II Telemetry Fusion

Fuse real-time On-Board Diagnostics (OBD-II) data — speed, brake pressure, steering angle — directly into the incident model for physics-grounded fault analysis.

Edge AI

Mobile Edge Deployment

Quantise models (GGUF INT4/INT8) for deployment on Snapdragon 8 Gen chips — enabling real-time in-car incident detection without any external hardware.

Thank You

Questions & Discussion

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Built with YOLO11 · MiniCPM-V · Mistral · Ollama